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ENG5059P: MSc Project (2022-23)

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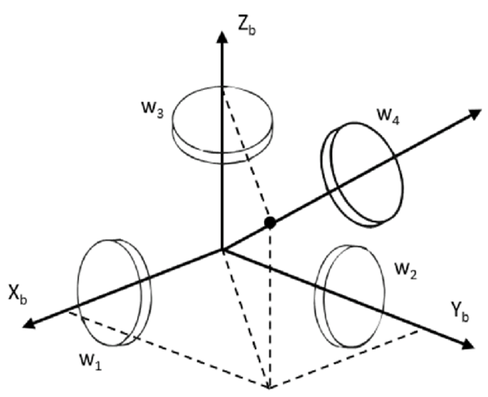
Deep reinforcement learning for attitude control of a satellite with faulty reaction wheels

Objective:

This project aims to investigate the robustness of DRL-based satellite attitude controller under fault scenarios, drawing conclusions about whether DRL controller can attenuate the impact of faults and remain control accuracy by validating the performance of the controller through simulation.

1. Introduction:

Deep reinforcement learning(DRL) has been used for a diverse set of applications including autonomous spacecraft attitude control. However, a relatively under explored area is the performance of these DRL-based controllers under scenarios where faults occur in the satellite system.



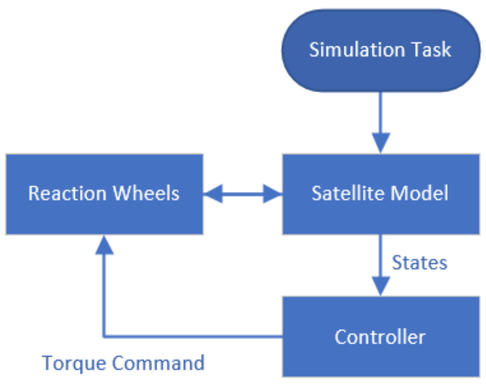
NASA 4 wheels configuration is commonly used in many modern spacecraft for its reliability. Typically, the attitude controller generates desired torques with respect to the axes and maps to available wheels. By means of mapping, if one wheel fails, the fourth wheel can compensate for it.

Regarding the DRL controller, it has been proven that, after training, the model has the capability to generate the correct wheel torque commands, thereby adjusting the attitude of the satellite. However, it has been trained to directly control the 4 wheels rather than mapping torques to them. In the case of a single wheel failure, will the DRL model retain its control capability?

2. Simulation Framework: Basilisk[1]

Basilisk is an astrodynamics simulation framework. It is extensively utilized for a variety of purposes, including flight data analysis and aiding in the development of AI-based spacecraft autonomy. It offers integrated systems modeling through convenient APIs, enhancing user interaction and functionality.

The torque of the reaction wheels is governed by a DRL controller. This is a trained model that issues command torques derived from the current state and the target attitude of the satellite. Concurrently, a PD controller has been developed for the purpose of model verification and validation.



3. Deep Reinforcement Learning:

DRL combines the concepts of deep learning and reinforcement learning, allowing an agent to learn how to behave in an environment by performing actions and simulating the results. It utilizes artificial neural networks to interpret complex data, enabling the agent to make optimal decisions even in complex scenarios.

Environment Definition

Fault injection time, reference attitude, and which wheel is set to be faulty are generated randomly. After a fault has been injected, the DRL model continues to generate torques for all wheels, but the torque of the faulty wheel will be set to 0 before dynamic simulation.

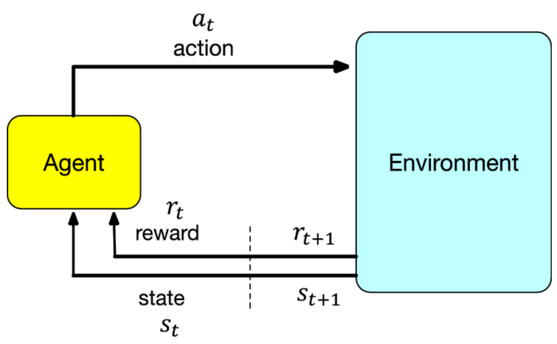
[1] 'Welcome to Basilisk: an Astrodynamics Simulation Framework — Basilisk 2.2.0 documentation'. https://hanspeterschaub.info/basilisk/index.html

Action Space:

The action of DRL agent is a 4 dimension vector, representing to the torques of reaction wheels. The range of torque is ±0.01 Nm.

Reward Function:

The agent will be rewarded when it reaches the target attitude or when the attitude error is reducing. The agent will be penalized when the angular velocity exceeds limitations. Two reward functions will be tested.

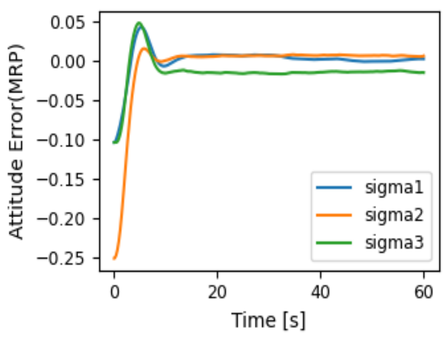


Training:

Each episode of training will simulate 6000 steps, where each step equals a 0.01 second dynamics simulation timestep, making each session equivalent to a real-world minute.

4. Results and discussion:

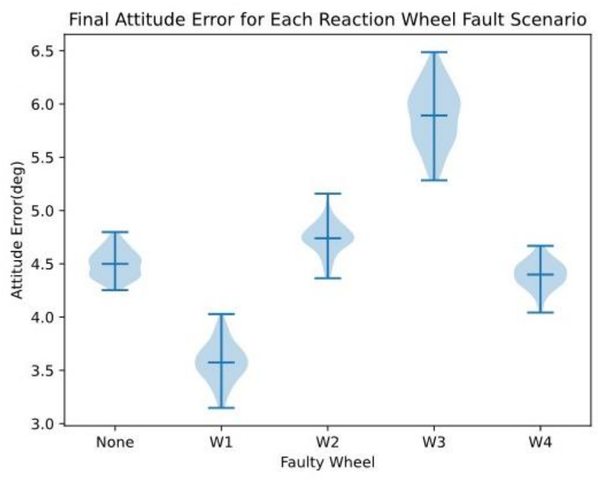
The first reward function did not meet expectations, as it was unable to effectively minimize the error. The second reward function is demonstrating promising results.



The plot provides an example in which a fault is injected at 30 seconds. Before the fault injection, the attitude error is adjusted to a relatively low value. Although the fault occurrence indeed impacts the performance of the DRL controller, the system still remains controllable.

Random Test: The trained model was evaluated using 500 groups of randomly generated data. The attitude at the 60-second mark was recorded, resulting in an average attitude error of 4.67 degrees.

Designed Test Cases: A series of combinations, involving target attitudes and fault conditions, has been designed to demonstrate the trained model's ability to adjust the satellite's attitude to any specified target under any fault condition. All test cases were successfully passed.



5. Conclusion:

The results demonstrate that the DRL controller can still achieve attitude adjustment objectives even with a wheel fault in the NASA 4-wheel configuration. In this project, the steady-state attitude error remains relatively large, and the response differs for each wheel fault. This is because the training is based on highly random fault conditions. However, it can be expected that with more training and parameter tuning, a controller with higher accuracy can be obtained.

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